

1. Research and explain how graphical user interfaces (GUIs) are used in everyday applications such as ATMs, e-commerce websites, or desktop tools like calculators. Give at least 2 examples that are not listed on the examples.

a) Microsoft Edge OS

Microsoft Edge provides a graphical user interface with menus, tabs, icons, and tools for browsing the internet. Users can navigate websites, open multiple tabs, use extensions, and access tools like a PDF reader all through visual elements without needing to type commands. Features like vertical tabs, split-screen mode, and 'Workspaces' help users manage their browsing experience more efficiently, while buttons and icons guide actions such as downloading files or bookmarking pages.

b) Android Mobile Operating System

Android uses a touch-based graphical interface featuring home screens, app icons, widgets, and navigation buttons. Users interact with their smartphones or tablets using gestures like swiping and tapping, making tasks such as opening apps, checking notifications, or changing settings quick and visual. The interface includes a status bar for alerts and quick settings, a home screen for app shortcuts and widgets, and a "Recents" screen for switching between apps. Android also supports split-screen mode and allows users to customize their layout by dragging apps or widgets as needed.

2. Explore Java's AWT or Swing library and identify key GUI components used in basic applications. In your answer, provide:

- At least 3 commonly used Java GUI components and their purpose.
- An example scenario or application where these components would be useful.
- Briefly describe how these components can improve user interaction.

a). JButton (Swing) / Button (AWT)

Represents a clickable button that can trigger specific actions when pressed. In a login application, a "Submit" button processes user credentials upon clicking. Provides a clear and interactive means for users to initiate actions, making the application more responsive and intuitive.

b.) JTextField (Swing) / TextField (AWT)

Allows users to input a single line of text. In a registration form, text fields are used for entering names, email addresses, or passwords. Facilitates data entry, enabling users to provide necessary information seamlessly.

c.) JLabel (Swing) / Label (AWT)

Displays a short string or an image icon, often used to describe other components. Labels next to text fields in a form indicate what information the user should enter ("Username:", "Password:"). Offers context and guidance, helping users understand the purpose of adjacent input fields, thereby improving usability.

References:

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